

Level: Bachelor

Semester: Spring

Year : 2017

Programme :BE

Full Marks: 100

Course: Social and Professional Issues in IT

Pass Marks: 45

Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) State major role of Von Neumann and Lady Ada in the field of computing. Explain about the history of Operating System. 7  
 b) Discuss in brief about the elements of a society. Explain about digital divide and governance of internet. 8
2. a) Define the terms ethics, moral and etiquette. Describe descriptive and Normative claims with example. 7  
 b) What is digital divide and how it can be minimized? Explain Health related issues for IT Professionals 4+4
3. a) Explain Engineering and computing as profession. Explain code of ethics of Nepal Engineering Council. 8  
 b) Define conflict of interest and whistleblowing. When is an engineer morally obligated to blow the whistle? 7
4. a) Explain about the Safety critical system. Define responsibility and accountability of computer users? Explain hardware design issues. 8  
 b) What do you mean by privacy risk? Explain about offensive speech and censorship in cyberspace. 7
5. a) What is offensive speech and censorship? Illustrate with relevant example for the importance of censorship in cyberspace. 7  
 b) What is the difference between trademarks, patents and copyrights? I often find good pictures or quotes on the Internet – can I use them in my own marketing materials? Does it make a difference if I make sure to give credit to the source? 8
6. Read the following case study and answer the questions below 15  
 Jan and Keith, both engineers, are faculty members at a major university. Both are seeking tenure from the university, and as part of the requirement,

they are required to publish original articles in scholarly and technical journals.

As a graduate student, Jan developed a paper that he had never published, but now feels would be an excellent topic to publish in a journal. Jan discusses this idea with Keith, and they agree to work together on revising the paper.

Jan does most of the work of revising the paper to bring it up to date. Keith's contributions are minimal, but Jan agrees to include Keith's name as co-author, so as to enhance Keith's chances of obtaining tenure. The article is accepted and later published in a scientific journal.

Questions:

- i. Is it ethically acceptable for Jan go back to his graduate work for an article to publish?
- ii. Should Jan's thesis supervisor be credited in some way, and if so, how?
- iii. How much should Jan and Keith have agreed upon at the start of their collaboration? Was it either unethical or unwise for Jan to include Keith's name as co-author?

2×5

Write short notes on: (Any two)

- a) IT policy of Nepal
- b) Phishing
- c) Pioneers of computing

Level: Bachelor

Programme: BE

Course: Social and Professional Issues in IT

POKHARA UNIVERSITY

Semester: Fall

Year : 2018

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) State major contributions of Jon Von Neumann in the field of computing. 7  
b) Identify and explain issues about technology and social change that often underlie discussions of the social implications of computer and information technology. 8
2. a) Define descriptive and normative claims. Explain utilitarianism and deontological theories. 7  
b) Define the term whistle blowing. What are the advantages and disadvantages of whistle blowing? Explain steps to solve whistle blowing in an organization. 8
3. a) Define values in design. Explain accuracy and democracy in internet. 7  
b) What is personal privacy? How have computers affected privacy? 8
4. a) What do you mean by denial of service? Explain different types of malicious programs. 7  
b) Define: 8
  - i) Copyright
  - ii) Trademark
  - iii) Patent
  - iv) Trade Secret
5. a) Describe in short IT Policy of Nepal. Do you believe that the Government of Nepal should do more work on the field of cyber law? If yes, give your suggestion. 7  
b) Define society. Explain impact of technology in society. 8

the following case study and answer the questions below:

In 1999, the National Telecommunications and Information Administration (NTIA), an agency of the U.S. Department of Commerce, using data from the Census Bureau, reported the following:

15

The good news is that Americans are more connected than ever before. Access to computers and the Internet has soared for people in all demographics groups and geographic locations. At the end of 1998, over 40% of American households owned computers, and one-quarter of all households had Internet access. Accompanying this good news, however, is the persistence of the digital divide between the information rich (such as Whites, Asians/Pacific Islanders, those with higher incomes, those more educated, and dual-parent households) and the information poor (such as those who are younger, those in rural areas or central cities). The 1998 data reveal significant disparities, including the following:

Households with incomes of \$75,000 and higher are more than twenty times more likely to have access to the Internet than those at the lowest income levels, and more than nine times as likely to have a computers at home.

Whites are more likely to have access to the Internet from home than Blacks or Hispanics have from any locations. Black and Hispanic households are approximately one-third as likely to have home Internet access as households of Asian/Pacific Islanders descent, and roughly two-fifths as likely as White households. Regardless of income level, Americans living in rural areas are lagging behind in Internet access. Indeed, at the lowest income levels, those in urban areas are more than twice as likely to have Internet access as those earning the same income in rural areas.

#### Questions:

- Why is access to the Internet important for democracy?
- "Author does not focus exclusively on the Internet; often he refers simply to access to the technology" why? Explain.
- What do you think will be the important ethical issues in the future of the Internet?

Write short notes on: (Any two)

- Digital Forensics
- History of Networking
- IEEE and ACM

now

2x5

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1.
  - a) Write an essay on prehistory of computing and programming.
  - b) "Information Technology has a major impact on poverty Alleviation and women empowerment" Do you agree or disagree? Justify with your opinion.
2.
  - a) What do you understand by digital divide and how can we bridge the digital divide? Explain.
  - b) Define computer Ethics. Explain three different ways of organizing ethical issues.
3.
  - a) Explain engineering and computing as a profession. List out the major responsibilities of an ICT Professional.
  - b) Write a note on Hacker ethics and netiquette.
4.
  - a) Suppose, you are a project manager in implementing a predictive modelling software to know the behaviors of teenagers. What could be possible risks and how will you resolve them.
  - b) Privacy can still be maintained even when the individuals are exposed to social networking sites. Do you agree or disagree? Justify your opinion.
5.
  - a) Cyber Crime has increased significantly with the increase in internet users. Elaborate this statement with reference to various types of crimes reported in Nepal.
  - b) What is fishing? Mention Types of Computer Crimes?
6.
  - a) What do you know about IT Related laws of Nepal? Explain highlighting the IT Policy of Nepal.

What do you understand by Intellectual Property Right? Discuss Trademark, patent, Design and Copyright. 8

Write short notes on: (Any Three)

- a) Code of Ethics of NEC 3×5
- b) Descriptive and Normative Claims
- c) E-Government
- d) Offensive speech and censorship in Cyberspace

POKHARA UNIVERSITY

Level: Bachelor

Programme: BE

Course: Social and Professional Issues in IT

Semester: Fall

Year : 2019

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

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*Attempt all the questions.*

1. a) Describe the contribution of Edward Dijkstra in the field of computing.  
b) How has society and technology affected each other? Discuss.
2. **Case: Facebook-related crimes on rise: Police** 7  
Kathmandu, Dec. 13: Cyber crimes arising from the misuse of social networking sites like Facebook are on the rise, police said Monday. Police investigation shows that cases of phishing, using fake Facebook accounts, email thieving, data hacking and blackmailing have increased. 8  
Lately, the police arrested two Nepali Facebook defrauders – a male and a female – on alleged charges of cheating people through Facebook by promising valuable gift items. Fraudsters – Anilnath Boki Shrestha of Sundhara, Lalipur district and Anita Maharjan of Banasthali, Kathmandu – were arrested for illegally taking money from innocent people. According to the police, Shrestha and Maharjan, posing themselves as gentlemen, promised the people valuable gifts on condition that they first deposited a small amount in fees in the said bank accounts to obtain the gifts. Nearly half a dozen victims had deposited their money in the bank accounts of the fraudsters after becoming a friend on Facebook. Superintendent of Police (SP) Meera Chaudhari of the Central Investigation Bureau (CIB) said that the Facebook defrauders had created fake Facebook identities. SP Chaudhari said that Shrestha was arrested from Thamel and Maharjan from her own home at Banasthali on Saturday. The swindlers had collected Rs. 500,000 from the victims, said the police. According to police statistics, over three to four years, the number of cybercrimes stood at 19 in 2013, which reached more than 130 by the end of 2016. There were 39 cyber crime cases in 2014 and 89 in 2015, the report stated. Two months ago, the Metropolitan Police Crime Division (MPCD) had arrested Tul Bahadur Tamang of Mankha VDC of Sindhupalchowk district for swindling Rs. 3.1 million from various people with promises of jobs in the USA and Canada. The CIB said that it had

registered more than 90 cases of Facebook cheating that include financial and Facebook blackmailing. Cases of cyber crime are increasing fast with each passing year along with the rise in the number of internet and social media users.

The Nepal Police statistics show that the number of cyber crimes surged by 105 per cent during the last fiscal 2014-15. Prakash Jung Karki, spokesperson at the Metropolitan Police Crime Division of the Nepal Police, said that there was a need for a comprehensive cyber law, establishment of a Nepali Computer Incident Response Team (CIRT) and capacity building of all the stakeholders to combat cyber-crime. Till date, the Nepal Police have been handling cases related to cyber-crime under the Electronic Transaction Act 2006.

- a) Taking the reference of the given case, discuss the current status of issues related to IT in Nepalese society. Relate ethical issues with the scenario. 10
  - b) What should be the role of professionals in dealing with the aforementioned condition? 5
3. Read the following case study and answer the questions below: 15

Many losses of skills are unintended side effects of computer but Microsoft made a conscious decision that has the effect of diminishing vocabulary and the richness and humor of languages. The thesaurus in Microsoft Word 2000 lists "trick" as the only synonym for "fool". It omits "clown", "blockhead", "idiot", "ninny", "dunderhead", "ignoramus" and others, all present in earlier versions.

Because of the popularity of Word and the ease of using its reference utilities, fewer people will consult standard references such as dictionaries and Roget's Thesaurus (which also contains "dupe" and "simpleton", for example).

Microsoft said that it eliminated words "that may have offensive uses".

#### Questions:

- a) Do producers of widely used reference works have an ethical responsibility to report the substance of their field accurately, or a social responsibility to remove potentially offensive words from the language?
- b) Should software engineers concern themselves with ethical issues any more than other individuals in society?

- c) Do computers and computer related technology really raise new ethical problems or do they just present old issues in a new form?
4. a) "Software risk is considered sensitive in case of life critical systems as the developer is the only person to blame if an accident occurs". Do you agree or disagree? Justify your opinion.
- b) Is it possible to enforce censorship in cyberspace? Justify your opinion with relevant example.
5. a) IPR (Intellectual Property Rights) have become a serious concern with the advancement in computing technology. What are some of the intellectual properties? Describe.
- b) What do you mean by digital forensics? Explain different cybercrimes. How would you prevent your computer from crime? Discuss.
6. a) Some people think that technology has made life easier; others claim the opposite. What is your opinion on this issue?
- b) Cyber Terrorism has been a major issue in today's context. What do you think how to deal with this issue? Put your views.
7. Write short notes on: (Any two)
- a) Contributions of Mark Zuckerberg
- b) Safety in Critical Systems
- c) Use of E-government systems in Nepal

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*Attempt all the questions.*

1. a) What do you mean by computing? Describe how the computing changes with the evolution in computer technology. 7  
b) Discuss in brief about the elements of a society. Explain about digital divide and governance of internet. 8
2. a) State major role of Von Neumann and Lady Ada in the field of computing. Explain about the history of Operating System. 7  
b) What is digital divide and how it can be minimized? Explain Health related issues for IT Professionals 8
3. a) Define the terms conflict of interest and whistleblowing. Explain the code of ethics of Nepal Engineering Council. 7  
b) Government has issued Secure Password Practices. Describe its gist. Also put some light on objectives of IT policy of Nepal. 8
4. a) Why is it important to focus continuously in safety of critical systems? Explain value centered design. 7  
b) What do you mean by computer crimes and digital forgery? Describe current scenario and effect of online pornography on children and state best approaches to reduce cyber crime in cyber space. 8
5. a) Explain about the Safety critical system. Define responsibility and accountability of computer users? Explain hardware design issues. 7  
b) What is offensive speech and censorship? Illustrate with relevant example for the importance of censorship in cyberspace. 8
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**Questions:**

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2×5

7. Write short notes on: (Any two)

- a) IT policy of Nepal
- b) Trademark
- c) Online Harassment

# POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Programme: BE

Course: Social and Professional issues in IT

Year : 2020

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) Discuss the contributions of Konrad Zuse in the field of computing. 7  
b) What is digital divide and how can you minimize or mitigate digital divide? How can be technology used for poverty alleviation? 8
2. a) Society evolves with time, and technology changes with social demand. How has technology impacted society and vice versa? Explain. 8  
b) Define computing? Describe how the programming languages changes with the evolution in computer technology. 7
3. a) Describe the professional ethics for a software engineer. Explain descriptive and normative claims. 8  
b) What are professional rights and professional responsibilities? Explain in brief. Briefly explain how can we avoid whistleblowing? 7
4. a) Define the terms conflict of interest and whistleblowing? Explain values in design. 8  
b) Discuss the central theme of code of ethics of ACM. 7
5. a) What are the best ways to protect an individual's privacy in internet space? Explain different privacy risks in detail. 7  
b) Define the terms cyber crimes and digital forgery? Describe current scenario and effect of online pornography on children and state best approaches to reduce cyber crime in cyber space in the context of Nepal compared to rest of the world. 8
6. a) What are Intellectual Property and Intellectual Property Rights? List out the different types of IP and explain any two of them. 8  
b) Explain different acts related to IPR in Nepal. State in brief about Right to Information act and Electronic Transaction Act in Nepal. 7

7. Write short notes on: (Any two)

a) Cyber terrorism

b) Ethical Relativism

c) Accuracy vs. Democracy in the Internet